DO NOW — RECALL

Lesson 142 · Networks: LANs, WANs, Standards and Pro

Name: _____ Class: _____ Date: _____

1. Which ACID property keeps committed transaction changes after a crash?

2. How many bits are in one byte?

3. What data structure uses LIFO?

4. What is a trace table used for?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Durability.

2. Model answer

Eight.

3. Model answer

Stack.

4. Model answer

Tracking variable state changes as an algorithm runs.

By the end, you can…

• LAN connects local devices; WAN connects distant sites.

• Switches connect local devices; routers forward traffic between networks.

• Protocols and standards let different equipment communicate.

Vocabulary — know these terms

LAN — Network spanning a relatively small geographical area.

WAN — Network connecting networks over larger distances.

protocol — Agreed rules for communicating between systems.

standard — Shared specification that enables interoperability.

NEW LEARNING

Read, annotate and explain

Identify scope

LAN connects local devices; WAN connects distant sites.

Interconnect devices

Switches connect local devices; routers forward traffic between networks.

Use open rules

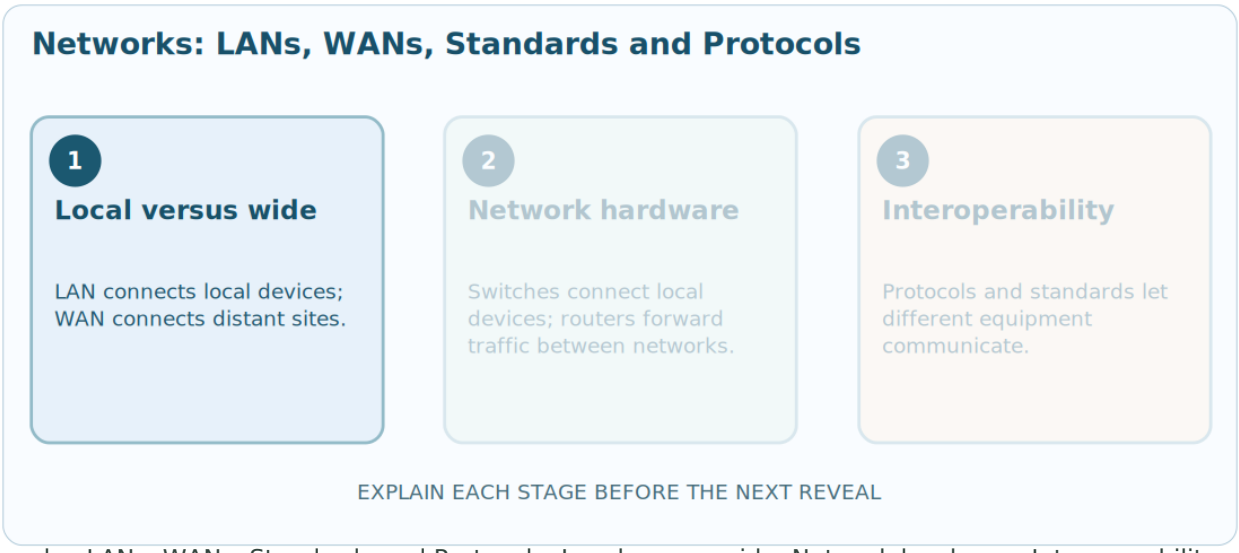
Protocols and standards let different equipment communicate.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

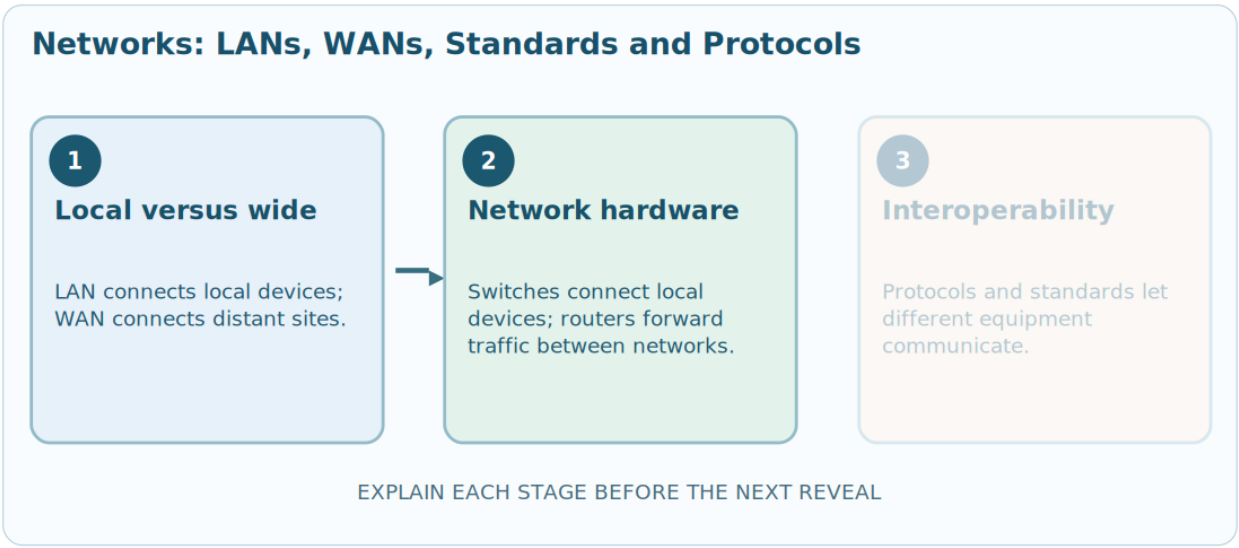
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Networks: LANs, WANs, Standards and Protocols: Local versus wide, Network hardware, Interoperability

New Learning

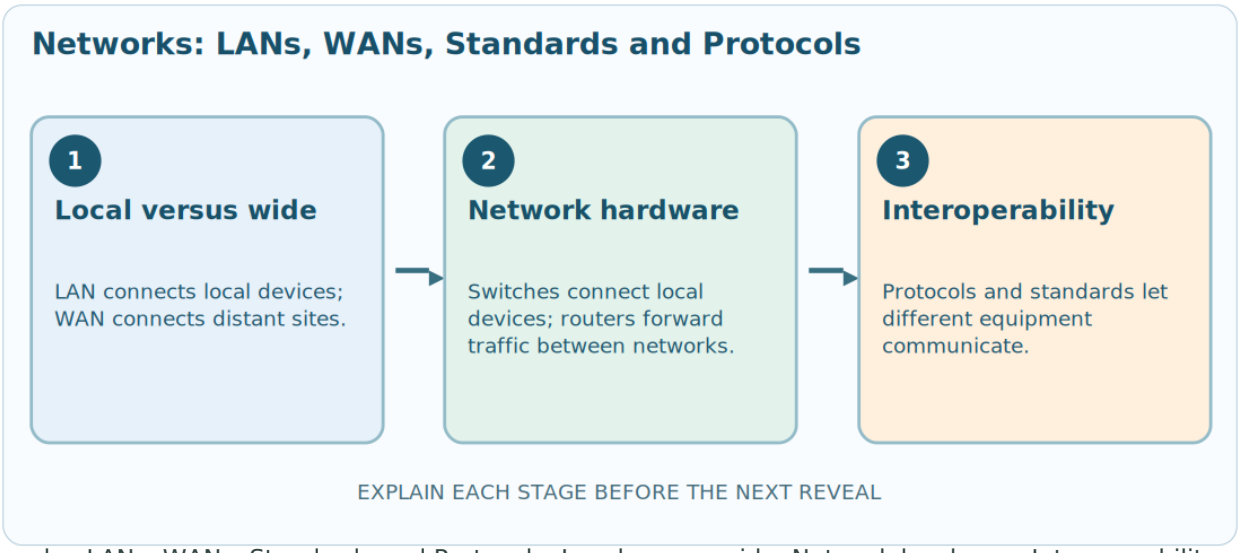


Networks: LANs, WANs, Standards and Protocols: Local versus wide, Network hardware, Interoperability

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Networks: LANs, WANs, Standards and Protocols: Local versus wide, Network hardware, Interoperability

I DO — WORKED EXAMPLE

Follow the worked method

Two classrooms share printers in the same school. Is this LAN or WAN, and why are protocols necessary?

Same school buildings

Identify the relevant information and make a prediction.

Local switched/Wi-Fi network

Explain the change using correct technical terms.

Standard protocols

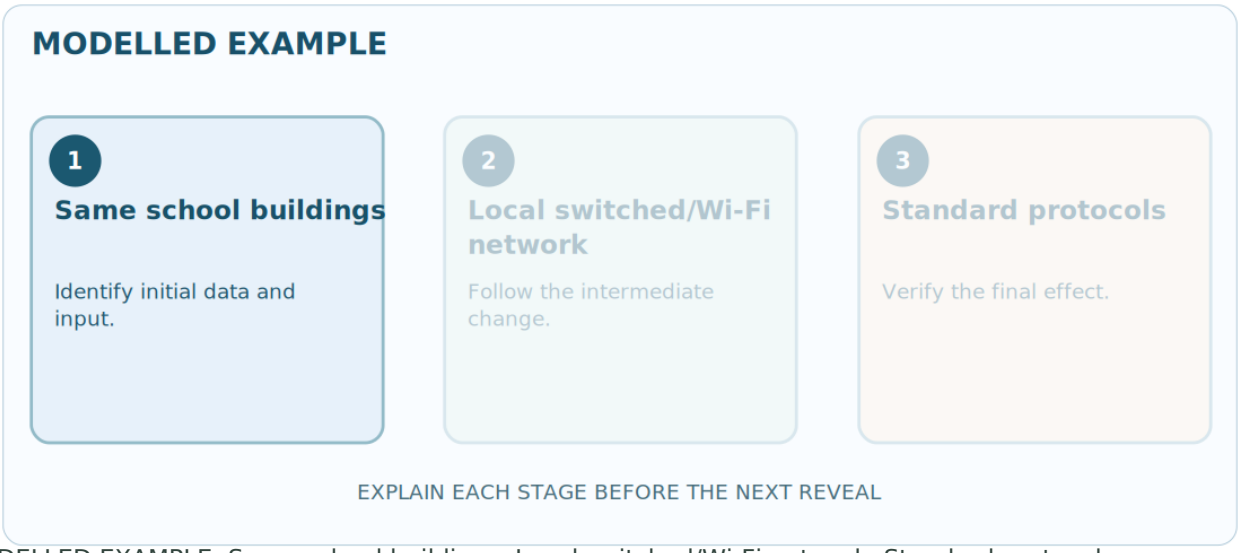
A school LAN connects local devices. Ethernet/Wi-Fi standards support compatible links and IP-based protocols allow data exchange. A router becomes relevant for communication beyond the local IP network.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Two classrooms share printers in the same school. Is this LAN or WAN, and why are protocols necessary?

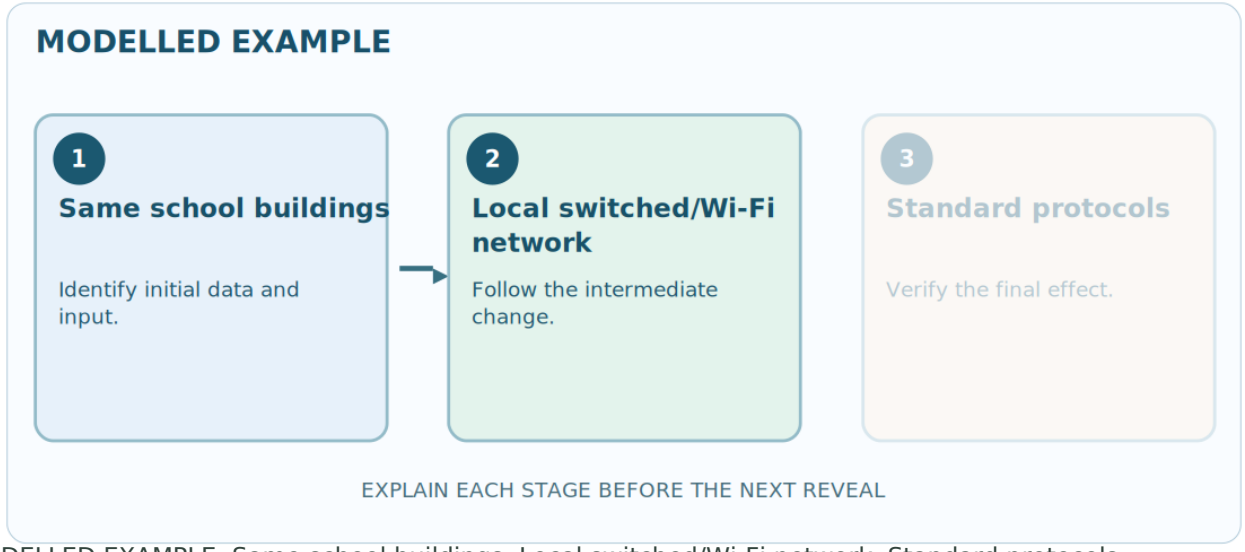
Identify the relevant information and make a prediction.



MODELLED EXAMPLE: Same school buildings, Local switched/Wi-Fi network, Standard protocols

Two classrooms share printers in the same school. Is this LAN or WAN, and why are protocols necessary?

Identify the relevant information and make a prediction.



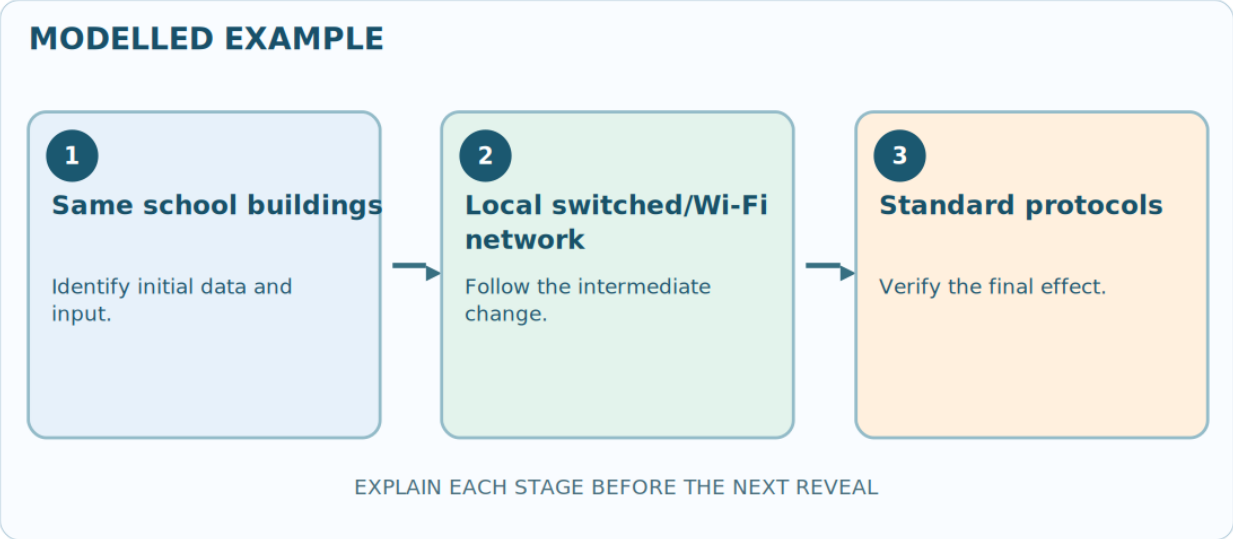
MODELLED EXAMPLE: Same school buildings, Local switched/Wi-Fi network, Standard protocols

I Do · original lesson diagrams

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MODELLED EXAMPLE: Same school buildings, Local switched/Wi-Fi network, Standard protocols

WE DO — GUIDED PRACTICE

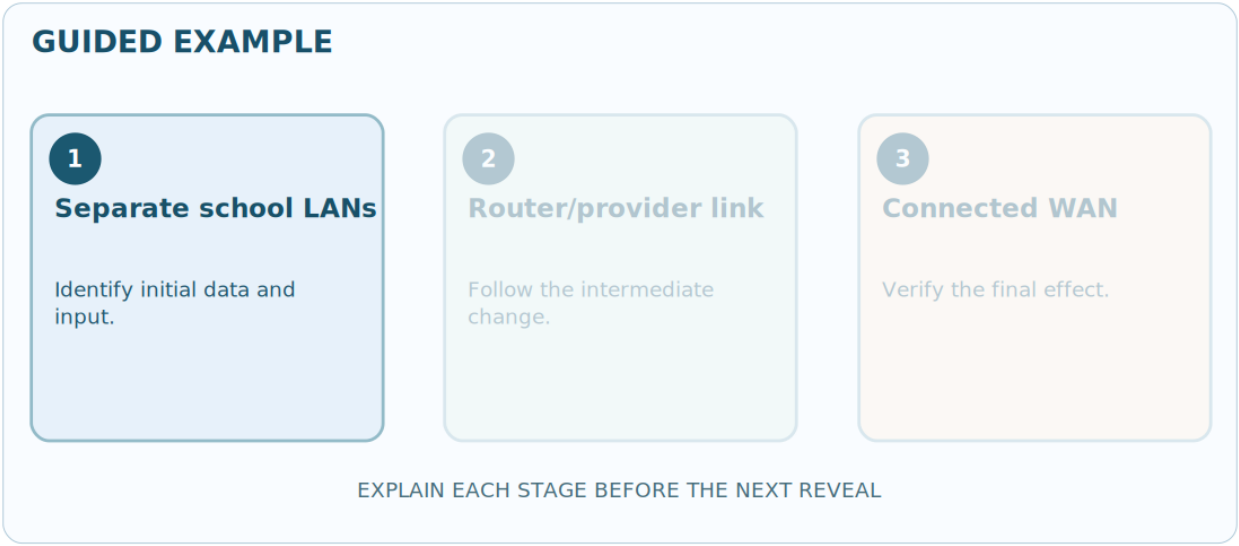
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A trust connects schools 40 miles apart using provider links. Classify the network.

Identify the relevant information and make a prediction.



GUIDED EXAMPLE: Separate school LANs, Router/provider link, Connected WAN

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Separate school LANs Identify the relevant information and make a prediction.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Differentiate LAN and WAN with examples. [6]

Question 2

Explain why protocols and standards matter when devices use different manufacturers. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A school requires a shared printer, wireless access and broadband internet. Explain suitable network devices and their functions. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A WAN is defined by having more devices than a LAN.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

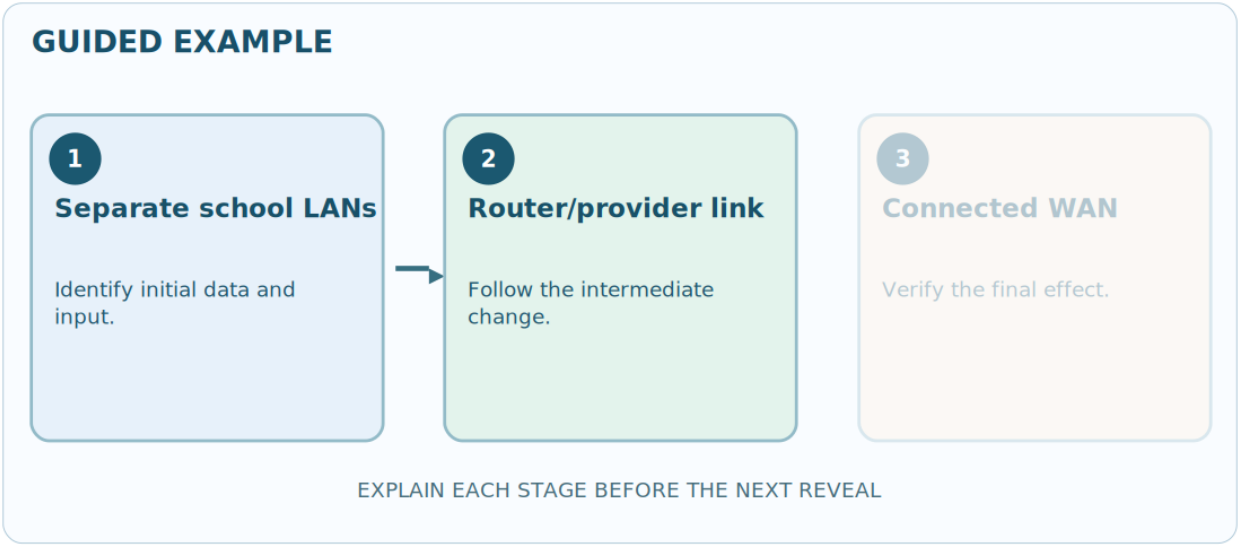
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

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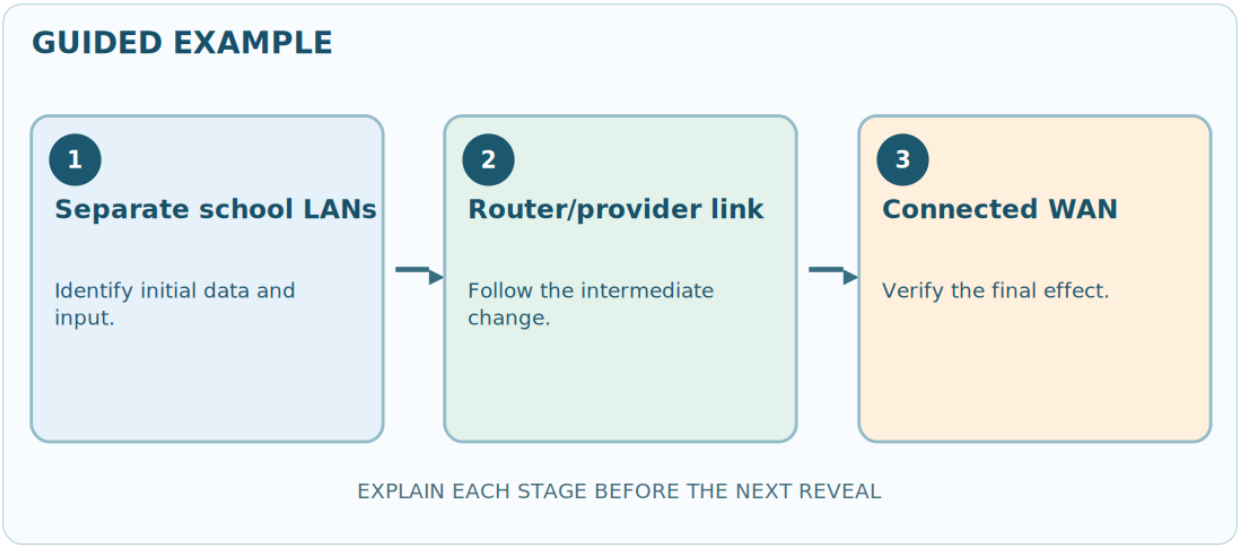
Identify the relevant information and make a prediction.



GUIDED EXAMPLE: Separate school LANs, Router/provider link, Connected WAN

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Identify the relevant information and make a prediction.



GUIDED EXAMPLE: Separate school LANs, Router/provider link, Connected WAN

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Router/provider link

Explain the change using correct technical terms.

Connected WAN

It is a WAN connecting multiple site LANs. Routers and service-provider infrastructure carry traffic between sites; secure connections and availability matter.

IMPROVE: Explain one correction to your own reasoning.